Limits and Continuity

What is a Limit?

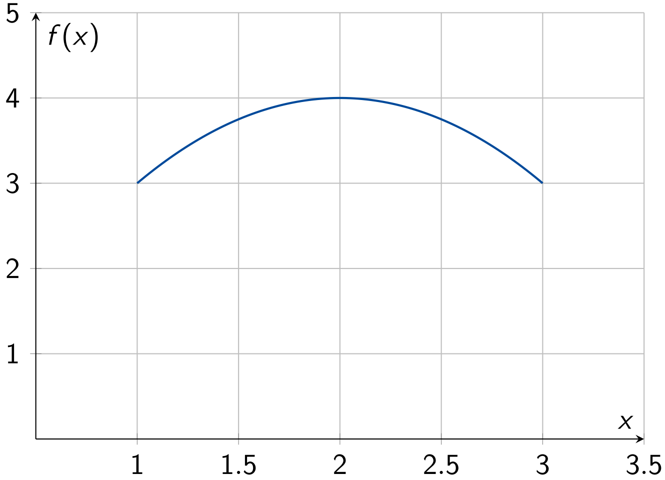
Informal Definition

The limit of a function $f(x)$ as x approaches a is the value that $f(x)$ gets closer and closer to as x gets closer and closer to a.

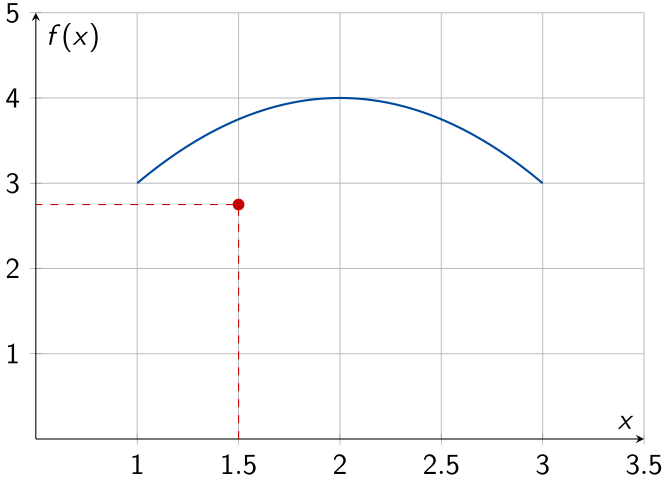
$$\lim_{x \to a} f(x) = L$$

- L is the limit value
- x approaches a but never equals a
- $f(x)$ approaches L as x approaches a

Visualizing Limits: Animation Concept

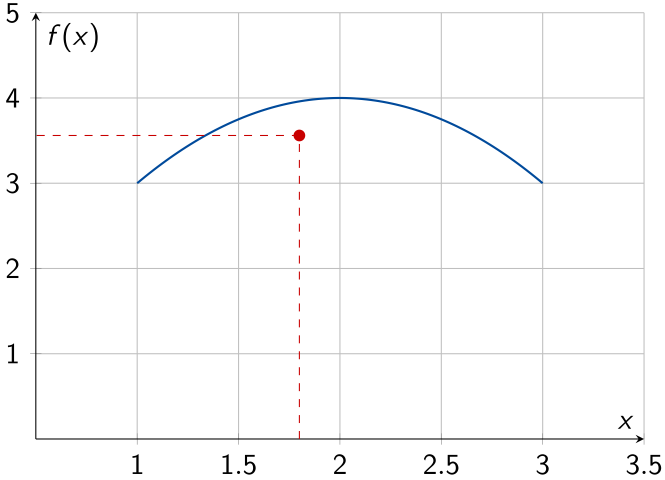


Visualizing Limits: Animation Concept



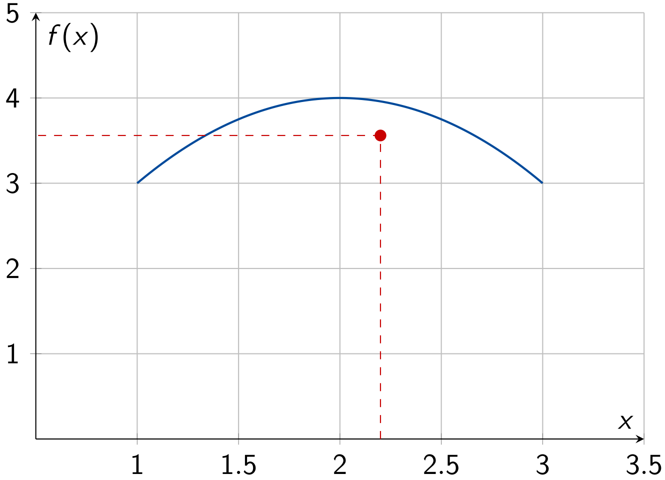
- As x approaches 2 from both sides...

Visualizing Limits: Animation Concept



- As x approaches 2 from both sides...
- $f(x)$ gets closer to 4

Visualizing Limits: Animation Concept



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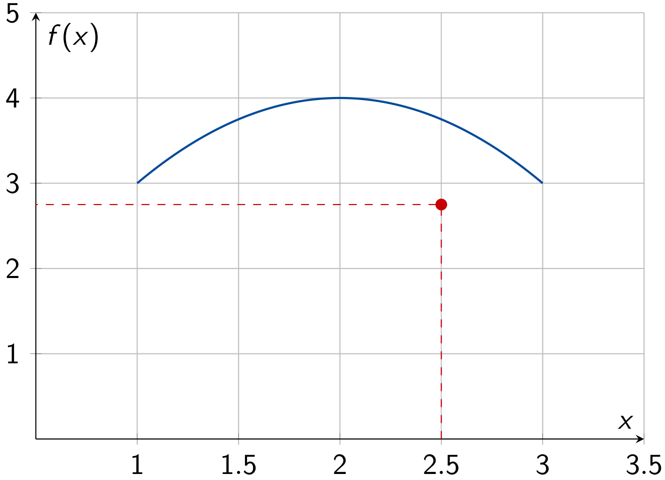

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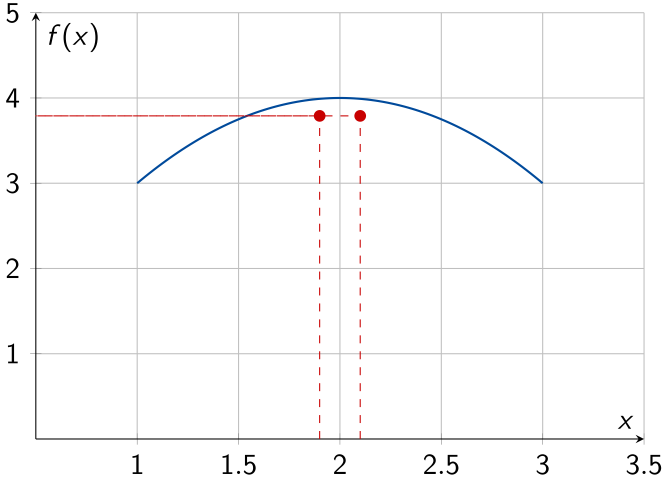
Visualizing Limits: Animation Concept



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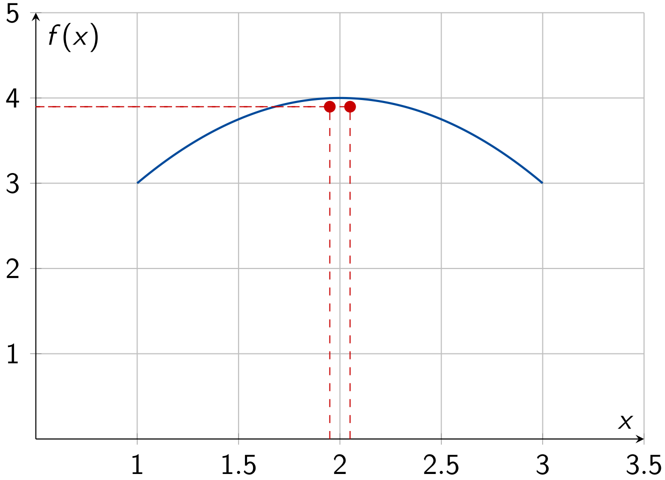

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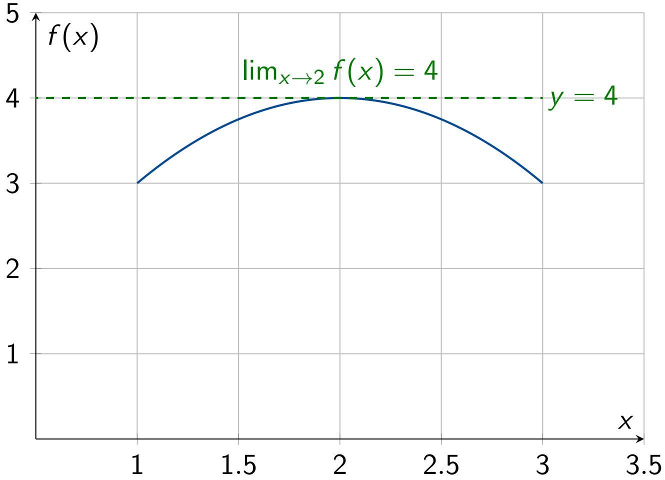
Visualizing Limits: Animation Concept



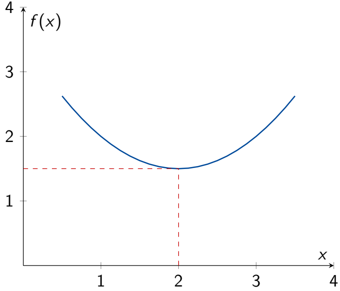
- As x approaches 2 from both sides...
- $f(x)$ gets closer to 4
- The limit is 4, even though $f(2) = 4$

The ϵ-δ Definition

Formal Definition

For every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$$



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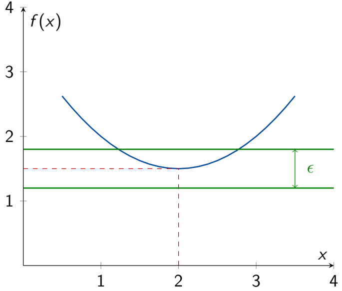

- ϵ: Maximum allowed error in $f(x)$

The ϵ-δ Definition

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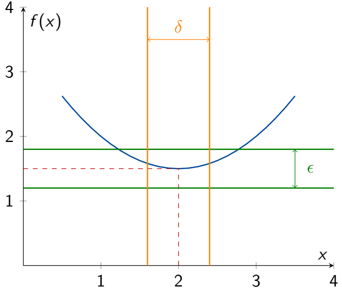

- ϵ: Maximum allowed error in $f(x)$
- δ: Corresponding tolerance in x

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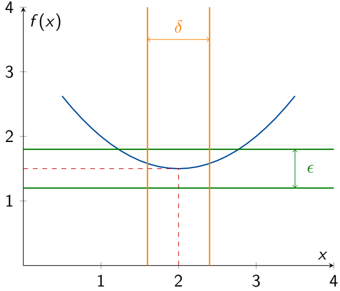


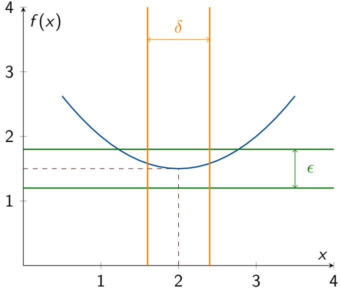
- ϵ: Maximum allowed error in $f(x)$
- δ: Corresponding tolerance in x
- Guarantees $f(x)$ stays within ϵ of L

The ϵ-δ Definition

Formal Definition

For every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$$



- ϵ: Maximum allowed error in $f(x)$
- δ: Corresponding tolerance in x
- Foundation for all calculus proofs

One-Sided Limits

- Left-hand limit: $\lim_{x \to a^-} f(x)$
- Right-hand limit: $\lim_{x \to a^+} f(x)$
- Limit exists at a if and only if both exist and are equal.

Example:

$$\lim_{x \to 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \to 0^-} \frac{1}{x} = -\infty.$$

Left-Hand and Right-Hand Limits

Left-Hand Limit

The limit of $f(x)$ as x approaches a from the left:

$$\lim_{x \to a^-} f(x) = L^-$$

Right-Hand Limit

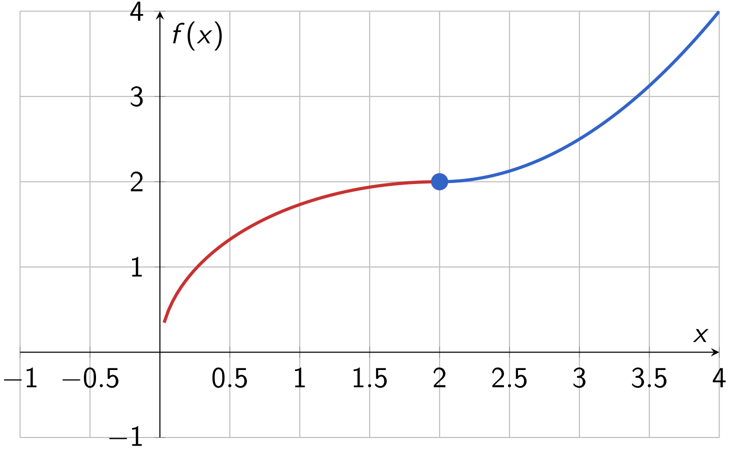
The limit of $f(x)$ as x approaches a from the right:

$$\lim_{x \to a^+} f(x) = L^+$$

Two-Sided Limit Exists If

$$\lim_{x \to a} f(x) = L \quad \text{exists if and only if} \quad \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

Approaching from Left and Right: Piecewise Function



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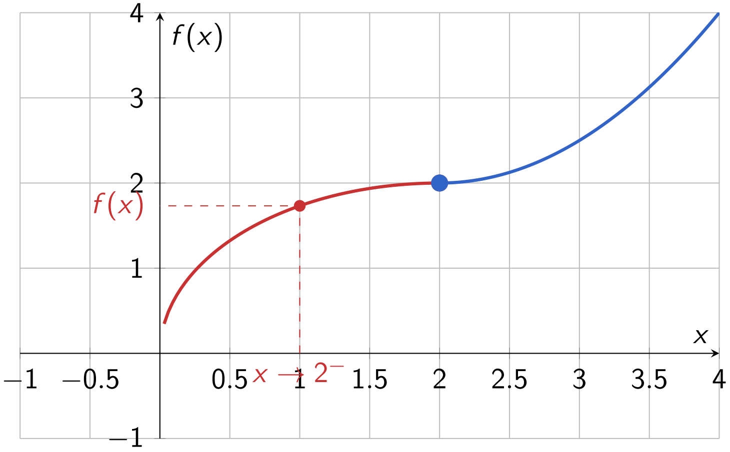
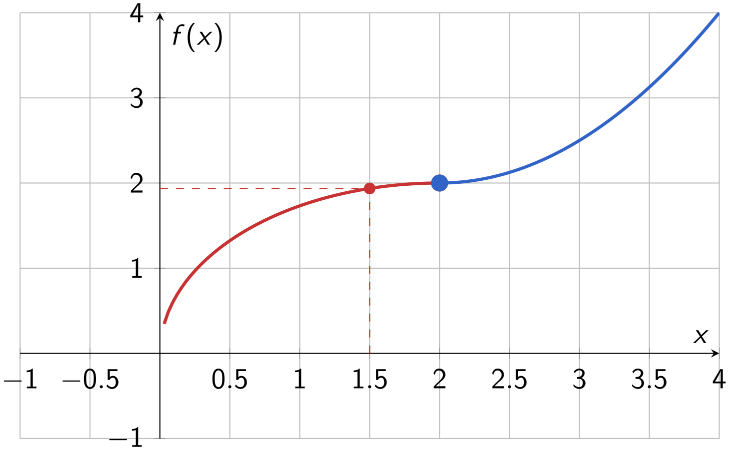


- Approaching from left: $x \to 2^{-}$

Approaching from Left and Right: Piecewise Function



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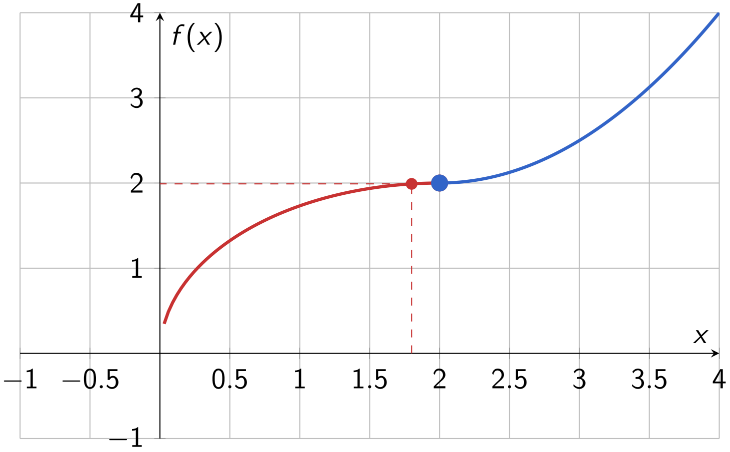

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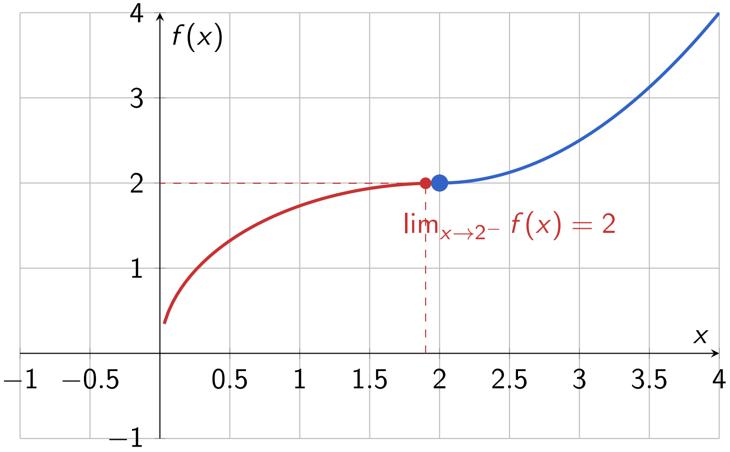
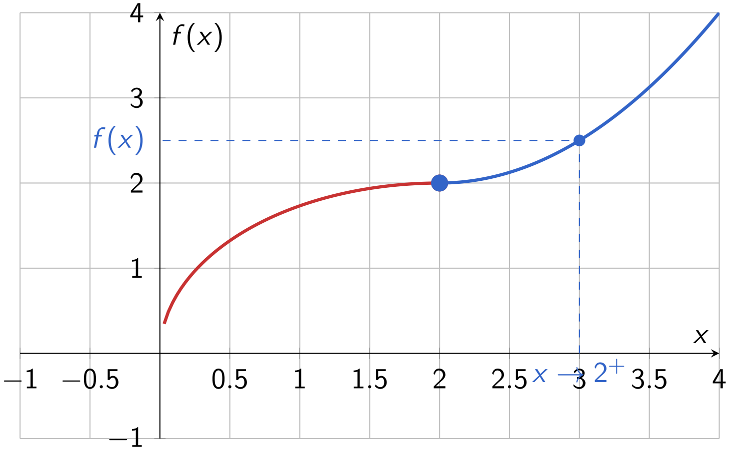


- Approaching from left: $x \to 2^-$
- Left-hand limit $= 2$

Approaching from Left and Right: Piecewise Function



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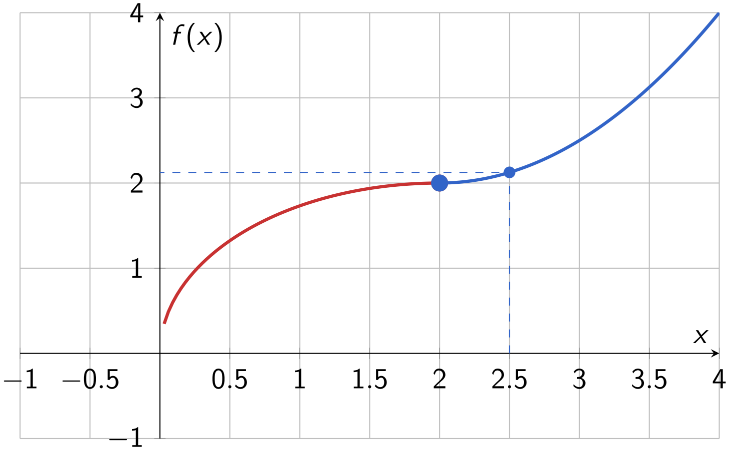

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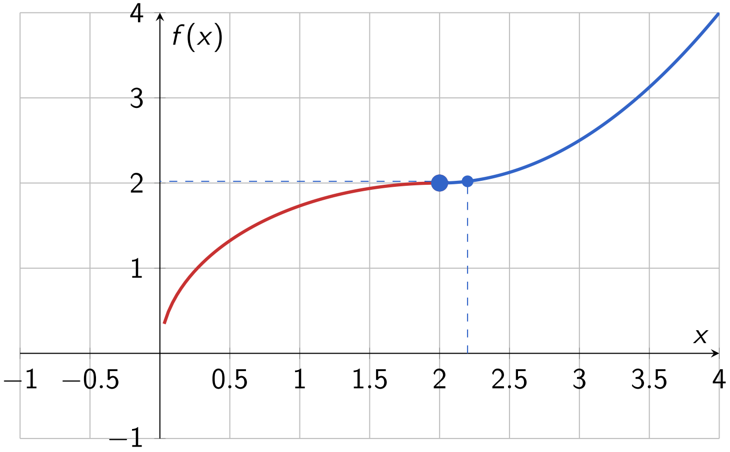

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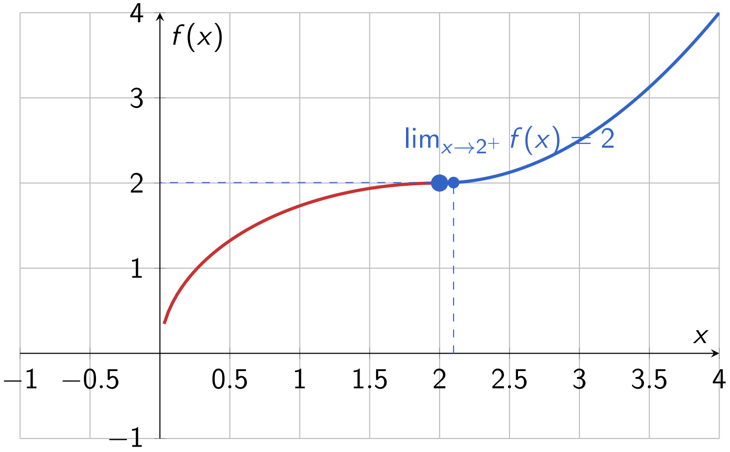
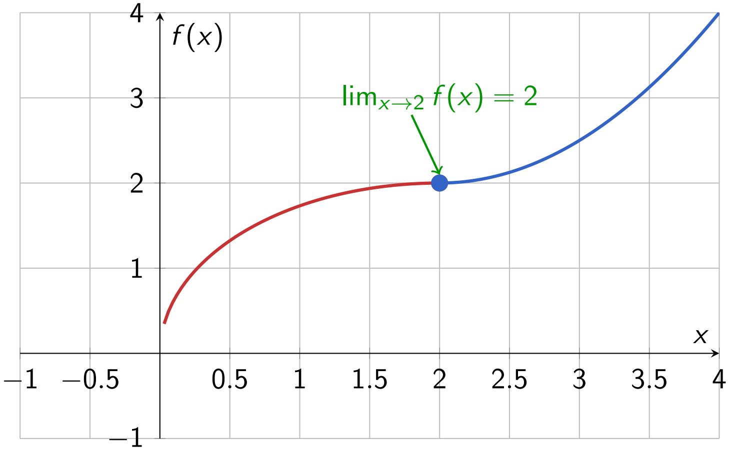
Approaching from Left and Right: Piecewise Function



- Approaching from left: $x \to 2^-$
- Approaching from right: $x \to 2^+$
- Right-hand limit $= 2$

Approaching from Left and Right: Piecewise Function



- Approaching from left: $x \to 2^-$
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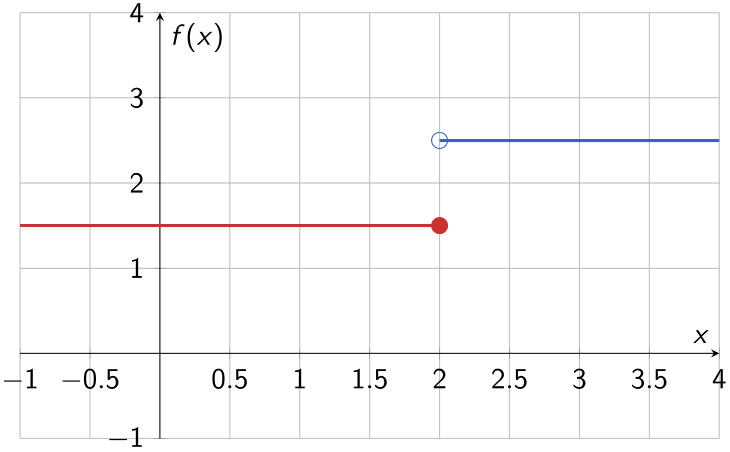
Function with Different Left and Right Limits



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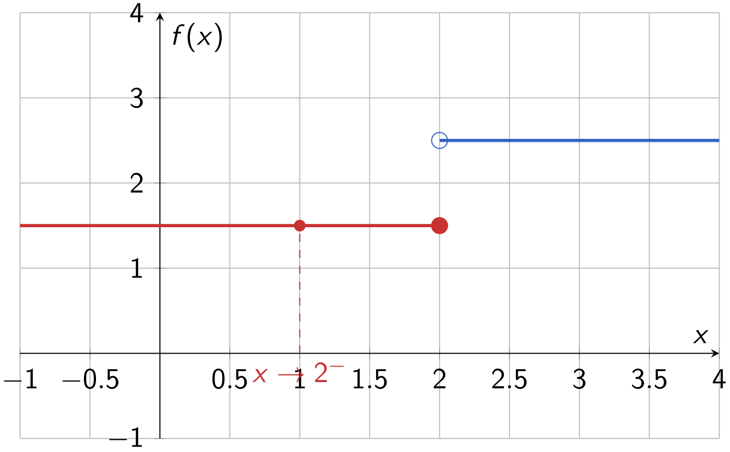
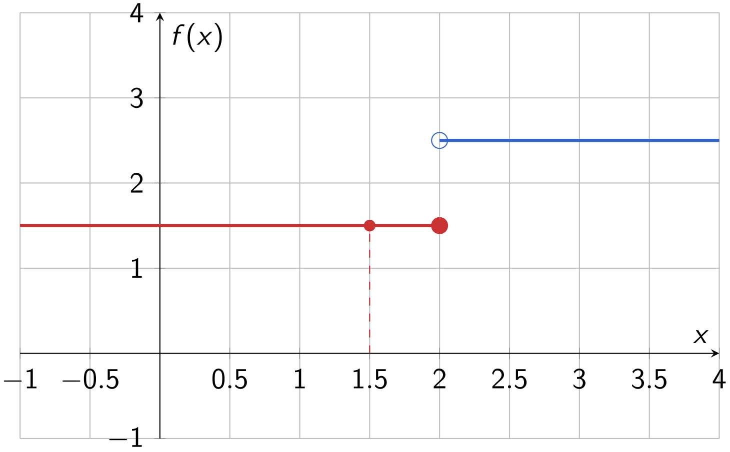


- Left-hand approach: $f(x) \to 1.5$

Function with Different Left and Right Limits



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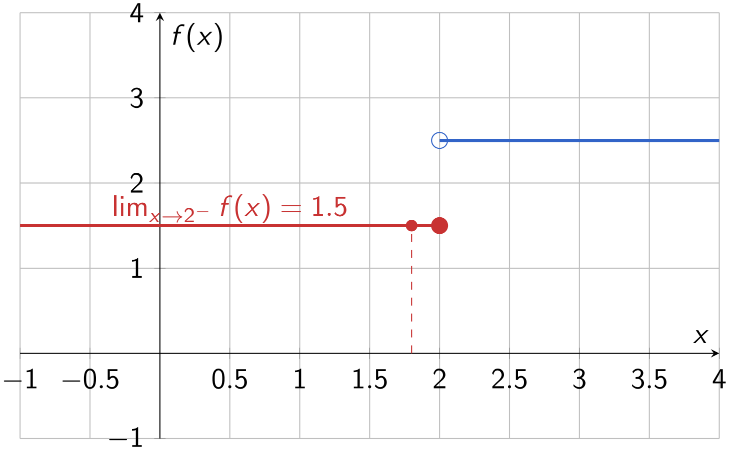
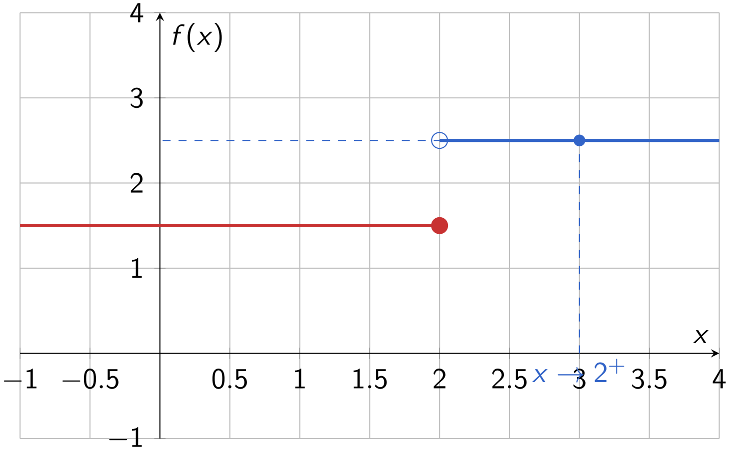


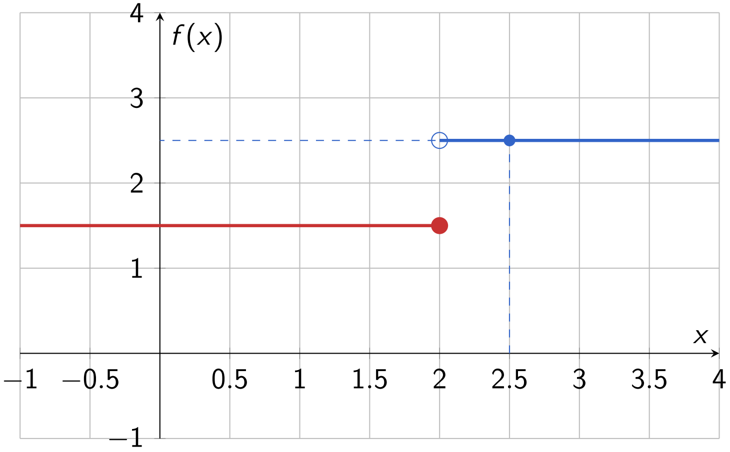
- Left-hand approach: $f(x) \to 1.5$
- Left-hand limit $= 1.5$

Function with Different Left and Right Limits



- Left-hand approach: $f(x) \to 1.5$
- Right-hand approach: $f(x) \to 2.5$

Function with Different Left and Right Limits



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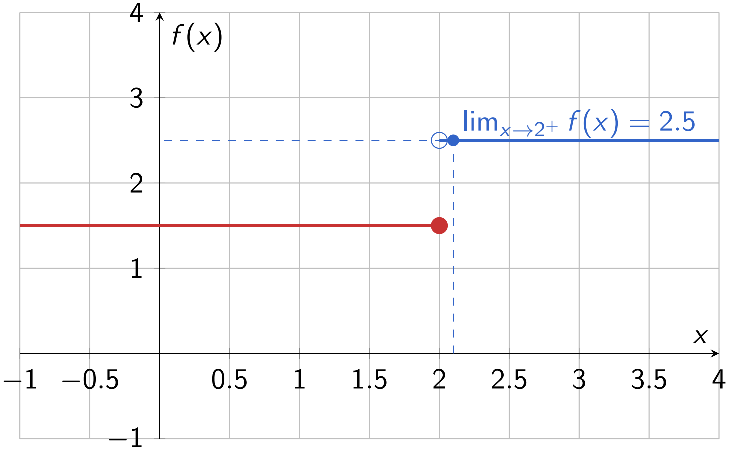
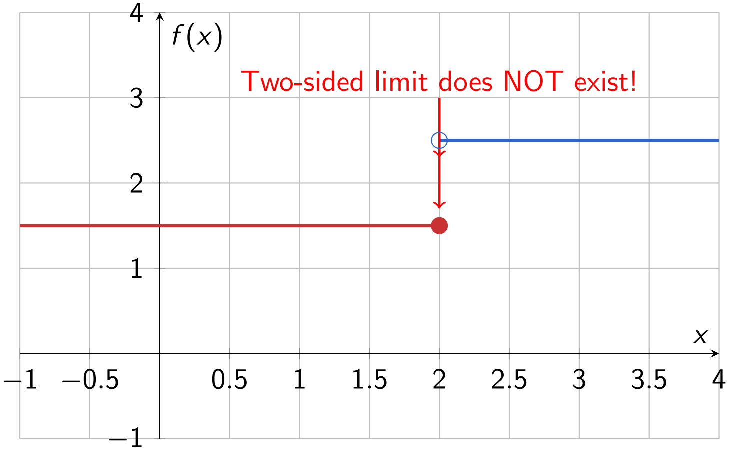
Function with Different Left and Right Limits



- Left-hand approach: $f(x) \to 1.5$
- Right-hand approach: $f(x) \to 2.5$
- Right-hand limit $= 2.5$

Function with Different Left and Right Limits



- Left-hand approach: $f(x) \to 1.5$
- Right-hand approach: $f(x) \to 2.5$

Formal Definition of One-Sided Limits

Left-Hand Limit (ϵ-δ Definition)

$$\lim_{x \to a^-} f(x) = L \quad \text{if for every } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that}$$

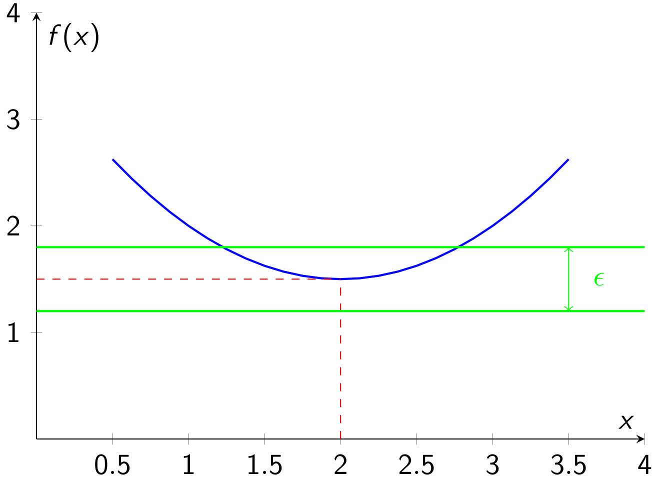
$$a - \delta < x < a \Rightarrow |f(x) - L| < \epsilon$$

Right-Hand Limit (ϵ-δ Definition)

$$\lim_{x \to a^+} f(x) = L \quad \text{if for every } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that}$$

$$a < x < a + \delta \Rightarrow |f(x) - L| < \epsilon$$

Visualizing the Formal Definition



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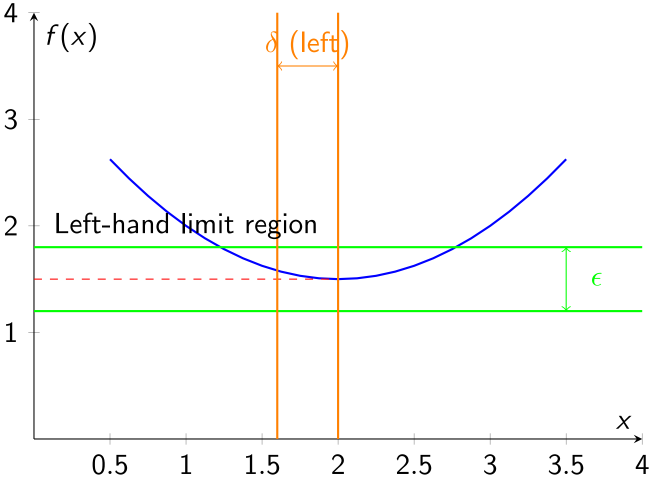
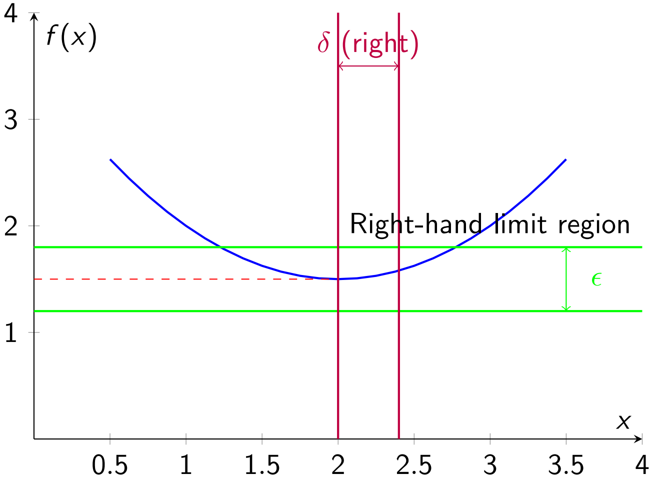


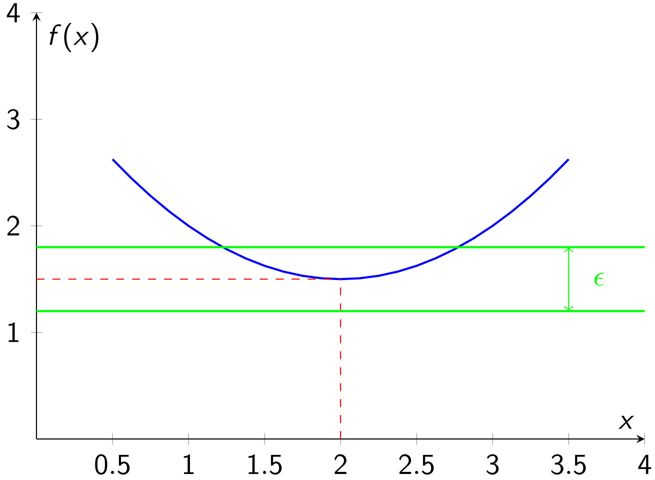
- Left-hand limit considers x values in $(a - \delta, a)$

Visualizing the Formal Definition



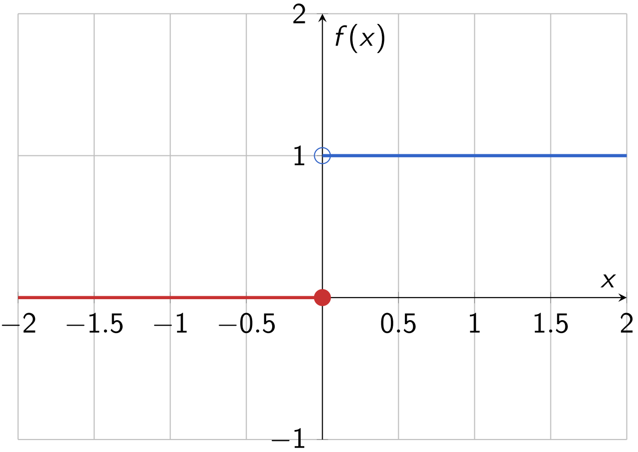
- Right-hand limit considers x values in $(a, a + \delta)$

Visualizing the Formal Definition

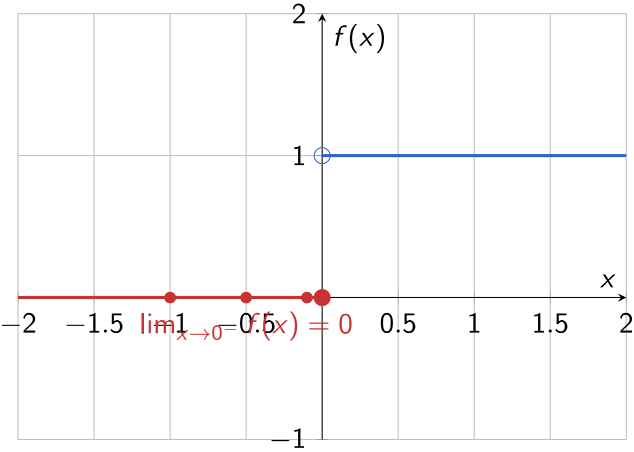


- Both must satisfy $|f(x) - L| < \epsilon$ in their respective intervals

Example: Step Function

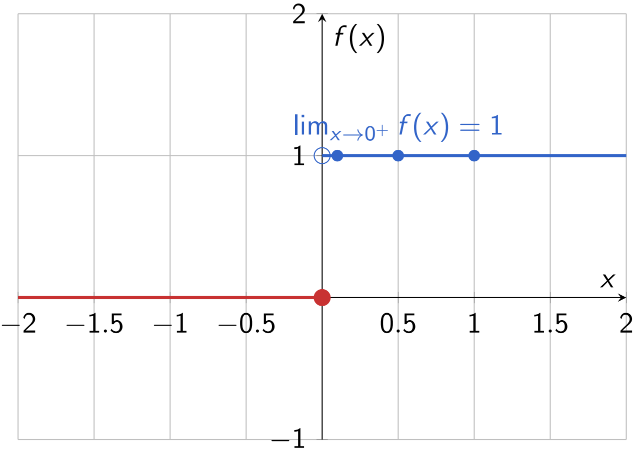


Example: Step Function



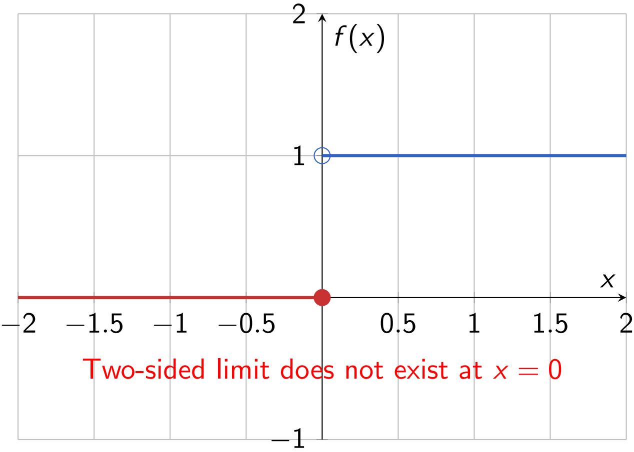
- Left-hand limit: $f(x) \to 0$ as $x \to 0^-$

Example: Step Function



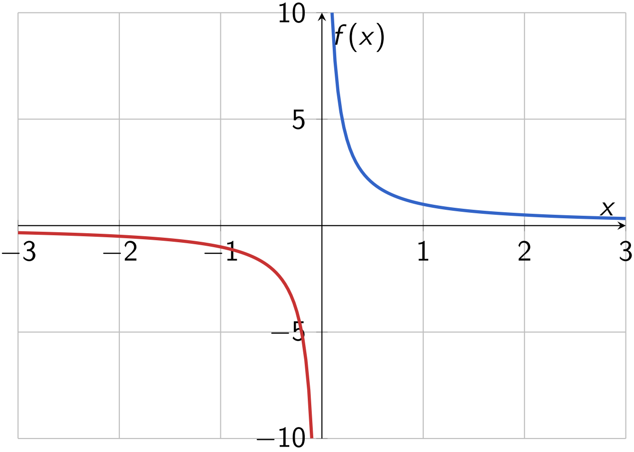
- Right-hand limit: $f(x) \to 1$ as $x \to 0^+$

Example: Step Function

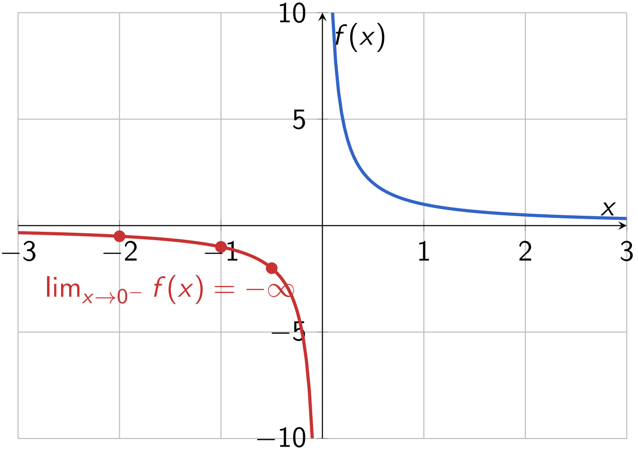


- Different values two-sided limit doesn't exist

Example: Function with Vertical Asymptote

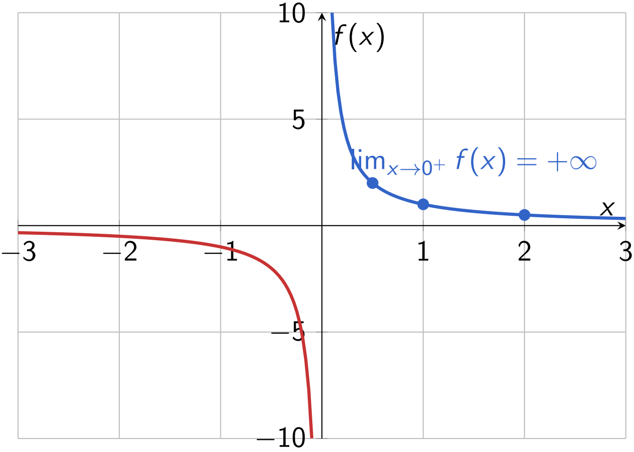


Example: Function with Vertical Asymptote



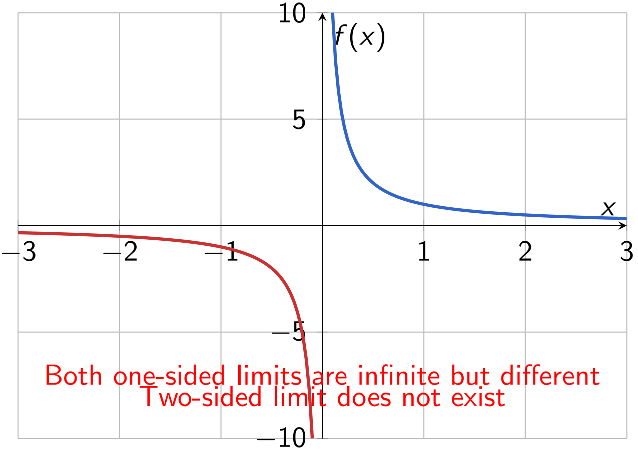
- Left-hand limit: $f(x) \to -\infty$ as $x \to 0^-$

Example: Function with Vertical Asymptote



- Right-hand limit: $f(x) \to +\infty$ as $x \to 0^+$

Example: Function with Vertical Asymptote



- Both are infinite but with different signs

Key Points About One-Sided Limits

- Left-hand limit: x approaches a from values less than a
- Right-hand limit: x approaches a from values greater than a
- Two-sided limit exists only if both one-sided limits exist and are equal
- Different one-sided limits indicate a discontinuity

Theorem

$$\lim_{x\to a} f(x) = L \quad \text{if and only if} \quad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$$

Infinite Limits

Definition

- $\lim_{x \to a} f(x) = \infty$ means: for every $M > 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies f(x) > M.$$

- Similarly defined for $-\infty$.

Example:

$$\lim_{x \to 0^+} \frac{1}{x^2} = +\infty.$$

Limits at Infinity

Definition

- $\lim_{x\to\infty} f(x) = L$ if for every $\varepsilon > 0$, there exists $M > 0$ such that

$$x > M \implies |f(x) - L| < \varepsilon.$$

Example:

$$\lim_{x\to\infty} \frac{3x+2}{2x+5} = \frac{3}{2}.$$